Letian Yang

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EDUCATION

University of Southern California - Los Angeles, CA, U.S. Aug.2023-May.2027

Ph.D. candidate, Department of Data Sciences and Operations, Marshall Business School; GPA: 4.00/4.00

Southern University of Science and Technology - Shenzhen, Guangdong, China Aug.2019-Jun.2023

B.S., Department of Statistics and Data Science; GPA: 3.78/4.00; National Scholarship (0.2%, Ministry of Education)

University of California, Berkeley - Berkeley, CA, U.S. Jan.2022-Jul.2022

Exchange Student, Berkeley International Study Program; GPA: 4.00/4.00

RESEARCH INTEREST

Causal Inference · Predictive Modeling · High-Dimensional Statistics

PROFESSIONAL EXPERIENCE

Graduate Researcher, Causal Effect Estimation via High-dimensional Linear Models Jan.2023-Present

- Developed theoretical guarantees for causal effect estimation in high-dimensional models and identified key failure modes of overparameterized inference, enabling stable business impact measurement in RCTs.
- Designed and analyzed a novel estimator (partially regularized OLS interpolator) that achieves unbiased causal effect estimation under high dimension, overcoming the limitations of traditional regularization methods (e.g., LASSO).
- Produced two manuscripts (one under review at AISTATS, another on arXiv), contributing new theoretical insights with applications in large-scale RCTs and observational studies.

Graduate Researcher, Freight Rate Prediction Jan.2024-Oct.2024

- Designed and launched an automated freight pricing engine that improved the efficiency of manual quote retrieval, greatly accelerated internal workflows, and enabled teams to handle a larger volume of customer requests.
- Unified internal shipment data with external freight price indices to build a richer, more robust statistical model supporting diverse global trade lanes and enabling more predictable, data-driven pricing decisions.
- Delivered a 12% improvement in forecast accuracy and faster quote responses, enhancing customer experience and supporting scalable growth across international routes.

Undergraduate Researcher, Gene Function Annotation with Deep Neural Networks Jan.2022-Aug.2023

- Developed a computational framework to refine Gene Ontology (GO) annotations in a tissue-specific and biological-process–specific context, addressing noise, missing labels, and inconsistency in existing databases.
- Applied deep neural networks, unsupervised clustering, and stability testing to extract robust gene–biological-process relationships from large-scale transcriptomic datasets, improving the interpretability of functional gene annotations.
- Delivered a scalable, data-driven approach that reduces manual curation effort, enabling more confident target prioritization, pathway analysis, and early-stage research decision-making.

Data Scientist Intern, Peking University Shenzhen Hospital Jun.2021-Aug.2021

- Identified 60 potential prognostic genes for downstream biomarker validation and precision oncology by analyzing TCGA and GEO head-and-neck tumor datasets; performed data cleaning, differential expression analysis, and survival modeling, enabling clinicians to prioritize relevant candidates for future translational studies.
- Designed and executed a patient oral-health survey, collecting 1,000+ responses; conducted age-stratified analysis (with focus on patients <15), applied descriptive statistics and clustered behavior profiling, and delivered findings that informed personalized clinical education strategies and prevention-focused care programs.

SKILLS SUMMARY

Programming Languages: R, Python, SQL, Java, MATLAB, LaTeX.

Systems and Tools: Linux, SSH, anaconda, AWS.

Data Science Libraries: NumPy, Scikit-learn, PyTorch, TensorFlow, NLTK.

Statistical Modeling: Regression and generalized linear models, Time series forecasting; Probability theory, Stochastic processes, Multiple testing; Convex Optimization, Linear Programming.

Causal Inference: A/B testing design, Synthetic Control, Double Machine Learning.